

Practical Demo: Bond Returns and Horizon Analysis

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This reading copy shows the calculations' results and figures. The HTML demo contains the full code in folded blocks and interactive scenario controls.

The desk question

A client buys a bond yielding 5%. **Will the client actually earn 5% over the next two years?** Build a horizon-value report that separates coupon income, reinvestment income, and the change in the bond's full price. Then explain which assumptions make a quoted yield an attainable compound return.

Purpose and assumptions

Use an illustrative five-year, 6% coupon bond with face value \$100. Coupons are semiannual. Buy immediately after a coupon at a 5% nominal annual YTM and sell immediately after the year-2 coupon. All cash flows are contractual; ignore default, calls, taxes, fees, and bid-ask spreads. Accrued interest is zero at both endpoints. Exit YTM and coupon reinvestment rates are independent scenario inputs, quoted as nominal annual rates compounded semiannually. No inputs are live quotes.

Read alongside [Lecture 2](#). Dollar results below are per \$100 face unless explicitly scaled to the client position.

1 Price the promised cash flows

With $n = 10$ half-year periods and $C = 100(0.06)/2 = 3$,

$$P_0 = \sum_{k=1}^{10} \frac{3}{(1 + 0.05/2)^k} + \frac{100}{(1 + 0.05/2)^{10}}.$$

Year	Payment	Discount factor	Present value
0.50000	3.00000	0.97561	2.92683
1.00000	3.00000	0.95181	2.85544
1.50000	3.00000	0.92860	2.78580
2.00000	3.00000	0.90595	2.71785
2.50000	3.00000	0.88385	2.65156
3.00000	3.00000	0.86230	2.58689
3.50000	3.00000	0.84127	2.52380
4.00000	3.00000	0.82075	2.46224
4.50000	3.00000	0.80073	2.40219
5.00000	103.00000	0.78120	80.46344

Purchase full price: 104.376032. The 6% coupon exceeds the 5% YTM, so the bond trades above par.

2 Distinguish quoted yields from realized returns

Current yield is annual coupon divided by today's price. It ignores both reinvestment and the gain or loss when the bond is sold or redeemed. YTM is the internal rate of return of promised payments at the purchase price. It is not a forecast of a short-horizon sale price.

	Percent
Measure	
Coupon rate	6.0000
Current yield	5.7484
Nominal annual YTM	5.0000
Effective annual YTM	5.0625

A 5% semiannual-compounding YTM corresponds to an effective annual rate of $(1.025)^2 - 1 = 5.0625\%$. Realizing that compound rate requires the relevant reinvestment and terminal-value assumptions, as the next calculation shows.

3 Build terminal wealth without double counting coupons

At $H = 2$ years, four coupons have been received. The sale price is ex-coupon, with six half-year payments remaining. If j is the annual reinvestment rate,

$$FV_C = 3 \left[(1 + j/2)^3 + (1 + j/2)^2 + (1 + j/2) + 1 \right],$$

$$W_H = P_H + FV_C, \quad HPR = \frac{W_H - P_0}{P_0}, \quad R_{\text{annual}} = \left(\frac{W_H}{P_0} \right)^{1/H} - 1.$$

The coupon paid at year 2 enters FV_C once and earns no reinvestment interest. The sale price excludes that coupon. This is a terminal-wealth return that assumes all coupon cash remains invested until the horizon.

Measure	5.0
Reinvest (%)	5.00000
Sale / principal	102.75406
Coupon cash	12.00000
Reinvestment income	0.45755
Price change	-1.62197
Terminal wealth	115.21161
HPR (%)	10.38129
Annual return (%)	5.06250

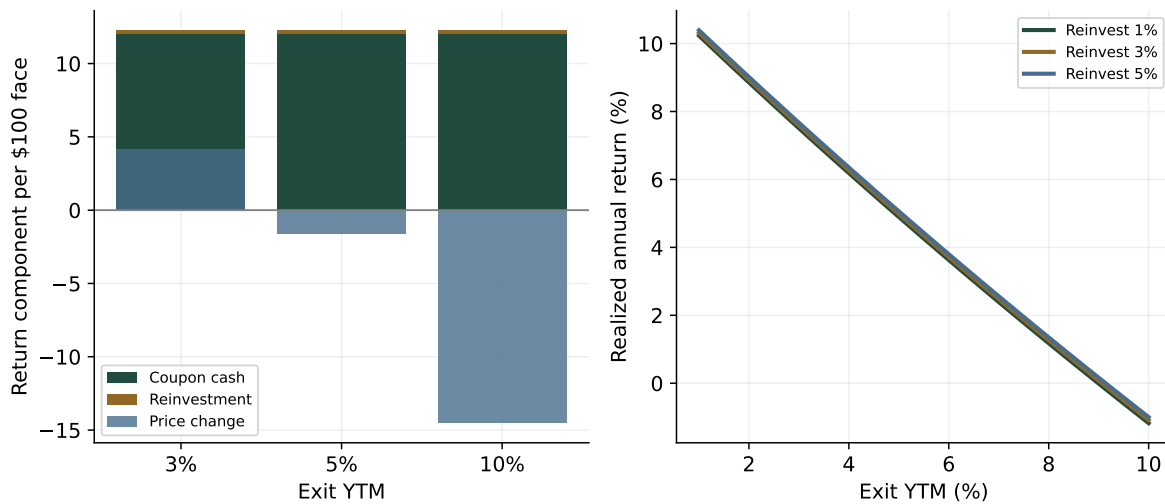
The unchanged-yield case includes a price decline toward par. Coupon income more than offsets that decline. A premium bond's coupon rate therefore overstates its return even when market yields do not change.

4 Separate price risk from reinvestment risk

Hold reinvestment at 3% while varying exit yields. Then hold the exit yield at 5% while varying reinvestment. This separation avoids assuming that sale yields and all past reinvestment rates move together.

Measure	3.0	5.0	10.0
Reinvest (%)	3.0000	3.0000	3.0000
Sale / principal	108.5458	102.7541	89.8486
Coupon cash	12.0000	12.0000	12.0000
Reinvestment income	0.2727	0.2727	0.2727
Price change	4.1697	-1.6220	-14.5274
Terminal wealth	120.8185	115.0268	102.1213
HPR (%)	15.7531	10.2042	-2.1602
Annual return (%)	7.5886	4.9782	-1.0860

Measure	5.0	5.0	5.0
Reinvest (%)	1.0000	3.0000	5.0000
Sale / principal	102.7541	102.7541	102.7541
Coupon cash	12.0000	12.0000	12.0000
Reinvestment income	0.0903	0.2727	0.4575
Price change	-1.6220	-1.6220	-1.6220
Terminal wealth	114.8444	115.0268	115.2116
HPR (%)	10.0294	10.2042	10.3813
Annual return (%)	4.8949	4.9782	5.0625



For this two-year horizon, a higher sale yield lowers the value of the remaining three-year bond. Higher reinvestment rates increase accumulated coupons. These are opposing effects with different magnitudes and timing.

5 Predict, change, explain

Predict whether a 100 bp increase in the **exit** yield raises or lowers realized return. Keep the reinvestment rate fixed at 3%. Move the HTML control, then explain why that change is not an increase in the return already earned.

The static scenario table above gives the corresponding PDF reference.

6 Validation checks

Passed: par pricing, return-component reconciliation, sale-price monotonicity, and the YTM reinvestment benchmark through maturity.

At maturity the exit yield no longer matters: principal is fixed. The accumulated coupon value still depends on reinvestment rates. Between coupon dates, use full purchase and sale prices, including accrued interest, and actual payment dates.

7 Mini Project

Project brief. Prepare a one-page return memo for a client buying \$2 million face of this bond and selling after two years. The client expects a 3% coupon reinvestment rate and considers exit YTM's of 3%, 5%, and 10%.

1. Report the cash purchase cost, terminal sale proceeds, coupons, reinvestment income, dollar profit, two-year HPR, and effective annual return for each case.
2. Explain why the 6% coupon and 5% purchase YTM differ from realized return.
3. Repeat the unchanged-exit-yield case with zero coupon reinvestment income.
4. Recommend whether the bond suits a client who cannot tolerate a two-year loss. State the model's limitations and identify a closer maturity match.

Worked solution: client return memo

Purchase cost is **\$2,087,520.64**. Coupons received are **\$240,000** over two years.

Exit YTM (%)	Sale / principal	Coupon cash
3.00	2170915.61	240000.00
5.00	2055081.25	240000.00
10.00	1796972.32	240000.00

Exit YTM (%)	Reinvestment income	Profit (\$)
3.00	5454.20	328849.18
5.00	5454.20	213014.82
10.00	5454.20	-45094.12

Exit YTM (%)	HPR (%)	Annual return (%)
3.0000	15.7531	7.5886
5.0000	10.2042	4.9782
10.0000	-2.1602	-1.0860

Reinvest (%)	Terminal wealth	HPR (%)	Annual return (%)
0.00000	114.75406	9.94293	4.85367

Interpretation and decision. The coupon rate determines contractual cash payments; it does not measure profit relative to the premium purchase price. The purchase YTM describes promised cash-flow discounting, while the horizon return also depends on sale price and reinvestment. The 10% exit scenario shows that coupon income need not prevent a loss. This five-year bond does not assure a positive two-year return. A default-free zero-coupon claim maturing at the client's horizon better matches a fixed nominal liability and eliminates coupon reinvestment risk. Its purchase price still determines its return; inflation, transaction costs, and selling before maturity remain relevant.